

SEAN MULLER

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EDUCATION

University of Miami

B.A. Mathematics | Minor in Political Science

B.S. Finance & Accounting (BSAF Program) | GPA: 3.41

Honors: Provost's Honor Roll, Dean's List

Selected Coursework: Advanced Calculus, Introduction to Probability, Statistical Analysis, Mathematical Finance, Machine Learning for Analytics, Data Analysis, Linear Algebra, Programming: Fundamentals & Algorithms

Coral Gables, FL

Expected Dec 2026

Completed 2026

EXPERIENCE

Independent Systematic Equity Research

May 2025 - Present

Independent | Systematic U.S. Equities | Quantitative Research

- Independently conceived, built, and operate an end-to-end systematic U.S. equities research and trading platform spanning data, hypothesis generation, model validation, portfolio construction, execution, risk, broker reconciliation, monitoring, and production release controls; now in the early stages of live capital management through Interactive Brokers following paper and staging validation.
- Built the research process from scratch across 500+ documented experiments and roughly 10 strategy families, using pre-specified hypotheses and pass/fail criteria, walk-forward and out-of-sample validation, placebo controls, bootstrap and cluster-robust inference, and multiple-testing deflation; rejected the majority of profitable-looking results before deployment.
- Treat strategy design as an evidence-driven engineering process: killed or redesigned components when simulation and paper evidence failed to support them, documented failure mechanisms, and promoted only behavior that survived independent replication, cost stress, launch-date sensitivity, and research-to-production parity checks.
- Engineered the data and production foundation supporting the research: 762M one-minute bars plus multi-horizon and options data across approximately 5,400 U.S. names, point-in-time reads and corporate-action handling, isolated research/dev/staging/production environments, pinned releases, and broker-state reconciliation.
- Built risk and operational controls with authority over trading, including exposure ceilings, drawdown governance, daily-loss and market-anomaly halts, correlation-spike liquidation, stale-data detection, bounded-chase execution, and independent trade-by-trade replication before production promotion.
- Designed an AI-native development workflow as an implementation multiplier, using specialized coding agents to accelerate research-to-production iteration while enforcing tests, ownership boundaries, and release gates; independently own the research hypotheses, statistical methodology, architecture, validation standards, risk design, and production decisions.

DE Consulting - Software Consultant

Jul 2026 - Aug 2026

Remote

- Built B2B SaaS operations platforms for commercial clients with \$100M+ annual revenue, translating executive and field workflows into production data models, APIs, analytics, and controls.
- Replaced fragmented Access, Smartsheet, Excel, and paper workflows with a centralized system of record; delivered a 95-model PostgreSQL schema, 451-endpoint API, 2,757-case CI test suite, and revenue/WIP/budget-variance analytics.

Earlier Experience: Nona Blue Modern Tavern - Host / Food Runner / Barback | May 2023 - Aug 2024

SKILLS

Quantitative Research: walk-forward/OOS validation, multiple-testing and deflated performance metrics, placebo/control design, bootstrap resampling, transaction-cost and fill simulation, point-in-time and survivorship-bias controls, time-series analysis, CAPM/factor diagnostics, Black-Scholes inversion and options Greeks

Programming & Data: Python, SQL, PostgreSQL, TimescaleDB, pandas, NumPy, SciPy, scikit-learn, statsmodels, pytest, mypy, Git

Systems & Tooling: FastAPI, Docker, Linux, systemd, TypeScript/React, Interactive Brokers API, AI agent orchestration (Claude Code), Excel